

semmcci: Staging

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```
library(semmcci)
library(lavaan)

# Data -----
data("Tal.Or", package = "psych")
df <- mice::ampute(Tal.Or)$amp

# Monte Carlo -----
## Fit Model in lavaan -----
model <- "
  reaction ~ cp * cond + b * pmi
  pmi ~ a * cond
  cond ~~ cond
  indirect := a * b
  direct := cp
  total := cp + (a * b)
"
fit <- sem(data = df, model = model, missing = "fiml")

## MC() -----
unstd <- MC(
  fit,
  R = 100L, # use a large value e.g., 20000L for actual research
  alpha = 0.05
)

## Standardized Monte Carlo -----
MCStd(unstd, alpha = 0.05)

#> Standardized Monte Carlo Confidence Intervals
#>
#>      est      se    R    2.5%  97.5%
#> cp      0.0906 0.0873 100 -0.0696 0.2238
#> b      0.4115 0.0859 100  0.2549 0.5623
#> a      0.1450 0.0930 100 -0.0468 0.3081
#> cond~~cond 1.0000 0.0000 100  1.0000 1.0000
#> reaction~~reaction 0.8116 0.0729 100  0.6333 0.9199
```

```

#> pmi~~pmi          0.9790 0.0266 100  0.9050 0.9998
#> indirect          0.4202 0.0442 100 -0.0228 0.1495
#> direct            4.1038 0.0873 100 -0.0696 0.2238
#> total             0.9160 0.0958 100 -0.0394 0.2938

# Monte Carlo (Multiple Imputation) -----
## Multiple Imputation -----
mi <- mice::mice(
  data = df,
  print = FALSE,
  m = 5L, # use a large value e.g., 100L for actual research,
  seed = 42
)

## Fit Model in lavaan -----
fit <- sem(data = df, model = model) # use default listwise deletion

## MCMI() -----
unstd <- MCMI(
  fit,
  mi = mi,
  R = 100L, # use a large value e.g., 20000L for actual research
  alpha = 0.05
)

## Standardized Monte Carlo -----
MCStd(unstd, alpha = 0.05)

#> Standardized Monte Carlo Confidence Intervals
#>
#>          est      se   R    2.5%  97.5%
#> cp          0.0939 0.0837 100 -0.0727 0.2659
#> b           0.3693 0.0783 100  0.2186 0.4979
#> a           0.1185 0.0868 100 -0.0581 0.2833
#> cond~~cond   1.0000 0.0000 100  1.0000 1.0000
#> reaction~~reaction 0.8466 0.0632 100  0.6946 0.9402
#> pmi~~pmi     0.9860 0.0237 100  0.9197 0.9997
#> indirect     0.0438 0.0377 100 -0.0219 0.1332
#> direct       0.0939 0.0837 100 -0.0727 0.2659
#> total        0.1377 0.0827 100 -0.0157 0.2996

```

References

- Pesigan, I. J. A., & Cheung, S. F. (2024). Monte Carlo confidence intervals for the indirect effect with missing data. *Behavior Research Methods*, 56(3), 1678–1696. <https://doi.org/10.3758/s13428-023-02114-4>
- R Core Team. (2026). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. Vienna, Austria. <https://www.R-project.org/>